

Solution Blueprint: Industrial Manufacturing Work Plan Assistant

Automated Work Plan Generation for Enterprise Industry PLM

Scalable Zero-Copy Lakehouse Analytics & Generative AI with Google Cloud Vertex AI

Executive Document Information

Attribute	Specification
Client / Domain:	Industrial Automation & Electronics Manufacturing (Enterprise Industry PLM Ecosystem)
Solution Category:	PLM & MES Workflow Automation (Manufacturing Work Plan Alternatives – WPA)
Technology Stack:	Google Agent Development Kit (ADK) 2.4.0, Gemini 3.5 Flash, Vertex AI Agent Engine, Google Cloud BigQuery, Gemini Enterprise
Data Architecture:	Apache Iceberg on AWS S3 / Cloud Storage → BigQuery BigLake External Tables (Zero-Copy)
Hosting Region:	Google Cloud europe-west3 (Frankfurt) / Data Region: EU
Document Classification:	Enterprise Solution Blueprint & Validated Reference Architecture
Status:	Validated Proof of Concept (PoC) & Production Architecture Pattern
Version:	1.0.0 (Anonymized Enterprise Distribution)

Executive Summary & Solution Highlights

Google Cloud Architecture Statement:
The **Industrial Work Plan Assistant ("Herbert")** demonstrates the symbiosis of **Google Cloud Generative AI (ADK + Gemini 3.5 Flash)** and **high-performance In-Database Data Analytics (BigQuery)** to automate mission-critical PLM workflows in enterprise manufacturing (Industry PLM). The underlying data lakehouse is hosted as open **Apache Iceberg tables**, queried in-place via **BigQuery BigLake External Tables** without data replication (Zero-Copy).

⚡ In-Database Speed	🎯 Mathematical Precision	🔒 Zero-Copy Data Lakehouse	🌐 Multi-Channel Integration
< 200 ms Query Latency across 35,000+ Work Plans	100% Deterministic Asymmetric Tversky Index	Open Apache Iceberg BigQuery External Tables	Gemini Enterprise & CLI Native Chat + REST endpoints

1. Problem Statement & Industrial Context

1.1 Challenges in Electronics & Industrial Assembly

In high-reliability industrial automation manufacturing (programmable logic controllers, distributed I/O modules, power supplies, and variable frequency drives), every new product release begins with an **Engineering Bill of Materials (E-BOM)** created by electrical and mechanical CAD engineers.

Manufacturing planners face the time-critical, manual task of generating a standardized **Work Plan Alternative (WPA)** within the corporate PLM system (Industry PLM). In major electronics plants, tens of thousands of historic, released work plans exist with highly specialized manufacturing operations (routing):

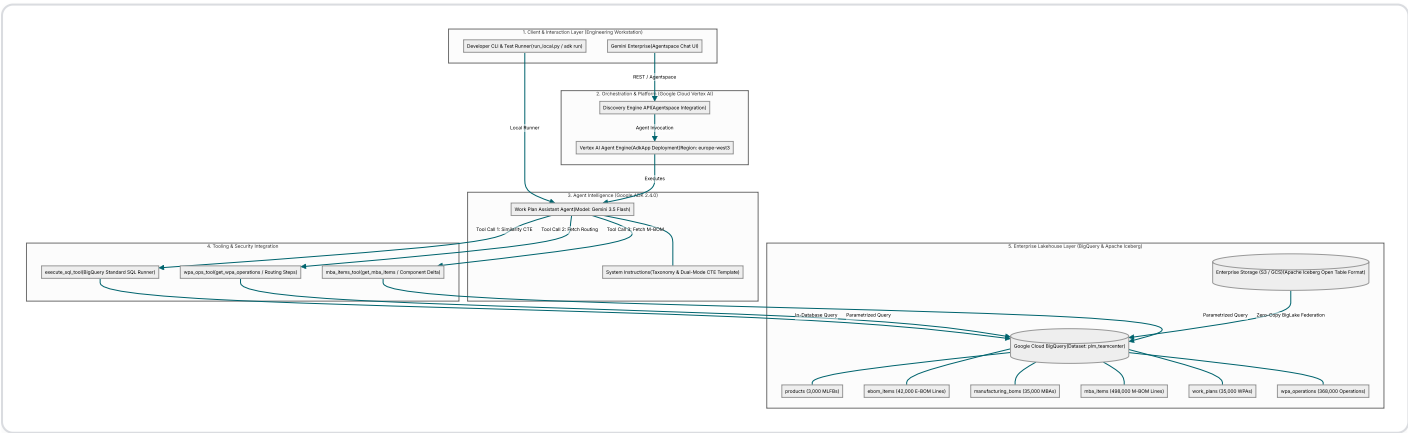
- **SMT Line (Surface-Mount Technology):** Solder paste printing, stencil cleaning, high-speed passive placement, and fine-pitch BGA/QFP processor placement.
- **Soldering Processes:** Multi-zone reflow soldering under nitrogen atmosphere (N_2) with profiled thermal gradients.
- **Inspection & Quality Gates:** 3D-AOI (Automated Optical Inspection) and AXI (Automated X-ray Inspection for void inspection under ball grids).
- **THT & Selective Soldering:** Through-Hole Technology for power connectors, relays, and electrolytic capacitors.
- **Testing Field:** In-Circuit Test (ICT needle-bed adapter), Boundary Scan, Flying Probe (APT), and firmware flashing.
- **Protection & Final Assembly:** Selective conformal coating, UV curing, automated housing screw fastening, thermal gap pad placement, and EMC shielding spring assembly.
- **End-of-Line (EoL):** Functional Test (FKT), high-voltage dielectric withstand test, and laser marking (2D DataMatrix serial codes).

1.2 Solution Objectives

The **Work Plan Assistant** automates and standardizes this routing discovery workflow:

1. **Multimodal & Unstructured Intake:** Parses new E-BOMs from structured JSON exports, CSV tables, or informal engineer free-text in English or German.
2. **Mathematically Formulated Similarity Search:** Executes a *3-Tier Asymmetric Tversky Multiset Index* directly in BigQuery across 35,000+ released work plans in milliseconds.
3. **Routing Extraction & Delta Analysis:** Recommends the Top-3 historical routing templates, extracts operational step sequences, and highlights part deltas (identical parts vs. new engineering items requiring process adjustments).
4. **Enterprise Integration:** Packaged with the **Google Agent Development Kit (ADK 2.4.0)**, deployed to **Vertex AI Agent Engine**, and exposed to operators via **Gemini Enterprise (Agentspace)**.
5. **Zero-Copy Lakehouse Federation:** Direct federated SQL access to **Apache Iceberg tables on S3 or GCS via BigQuery BigLake**, eliminating redundant ETL data movement.

2. End-to-End System Architecture



3. Data Model & Schema Design (PLM to BigQuery)

The relational schema mirrors the core entities of enterprise PLM systems:

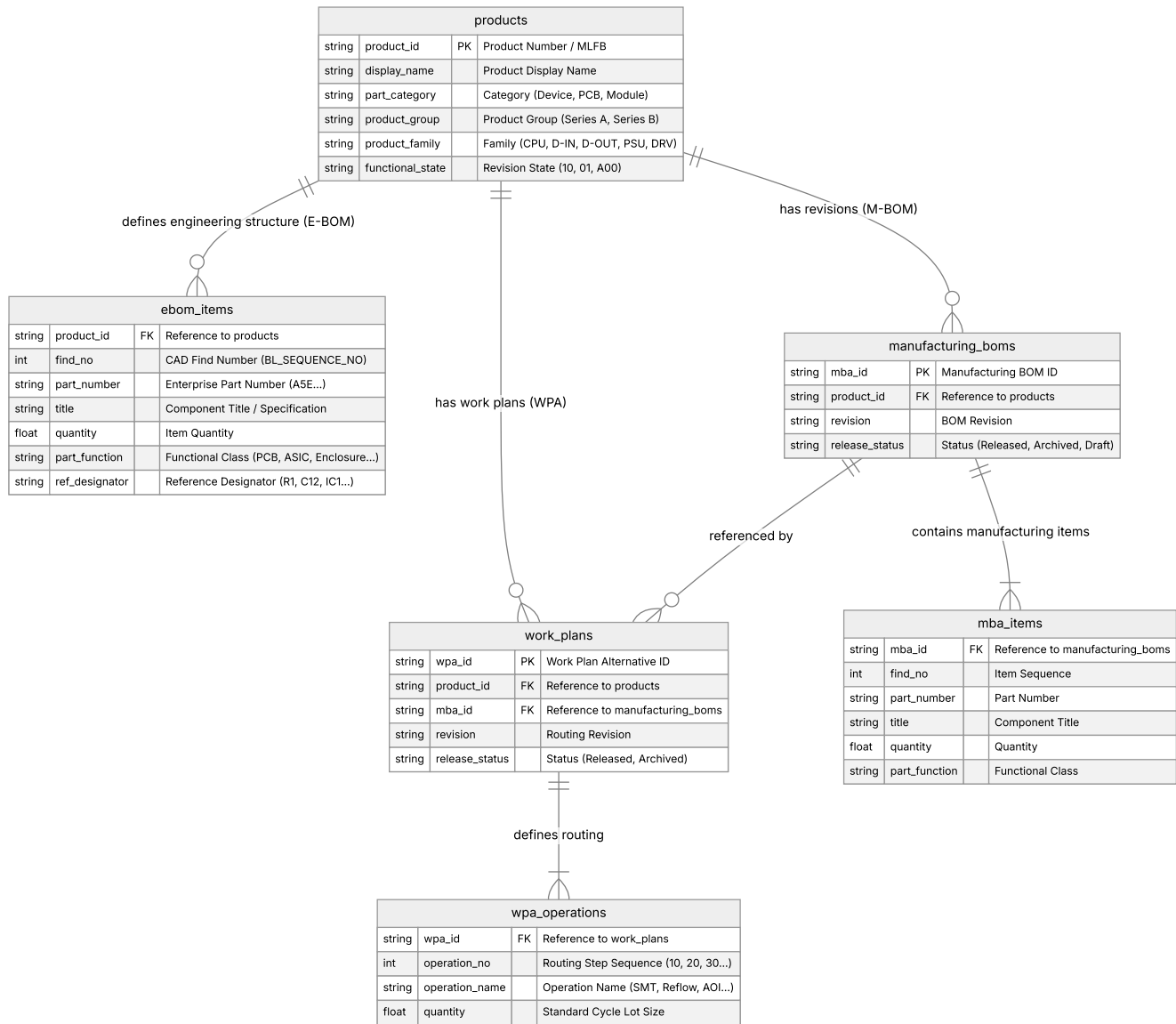


Table Specifications & Clustering Optimization

Table Name	Records	Partitioning / Clustering	Description
products	3,000	<code>['part_category', 'product_group', 'product_family']</code>	Master product catalog with categorization filters.
ebom_items	42,239	<code>['product_id', 'part_function']</code>	Engineering BOMs with designator coordinates.
manufacturing_boms	35,155	<code>['product_id', 'release_status']</code>	Manufacturing BOM revisions (exactly one active <code>Released</code>).
mba_items	498,317	<code>['mba_id', 'part_function']</code>	M-BOM lines mapped into 11 canonical functional classes.
work_plans	35,155	<code>['product_id', 'release_status']</code>	Work plan headers linked to corresponding released MBAs.
wpa_operations	368,513	<code>['wpa_id']</code>	Discrete routing steps (SMT, Reflow, AOI, THT, Coating, EoL).

4. Algorithmic Formulation: 3-Tier Asymmetric Tversky Index

4.1 Why Discrete Multisets Outperform Dense Vector Embeddings

Pure dense vector embeddings (Cosine Similarity over text embeddings) fail in manufacturing routing discovery for three fundamental reasons:

- 1. **Discrete Multiplicities & Tooling Dependencies:** A controller with 2x microcontrollers and 1x Ethernet PHY requires a different routing sequence than one with 1x microcontroller and 4x Optocouplers. Dense embeddings inappropriately smooth these discrete boundaries.
- 2. **Deterministic Auditability:** Plant manufacturing engineers cannot accept non-deterministic "black-box" scores. Every similarity score must be provably explainable in terms of exact part matches, missing specifications, and excess items.
- 3. **In-Database Execution:** Generating embeddings across millions of BOM positions introduces massive computational latency. In contrast, set-theoretic BigQuery SQL aggregates run across 35,000+ candidates in **under 200 ms**.

4.2 Mathematical Formulation

The similarity metric $S(Q,T)$ between the query BOM (Q) and a candidate template (T) is defined as an **asymmetric Tversky index over multisets**:

$$S(Q,T) = \frac{\text{Matched Mass}}{\text{Matched Mass} + \alpha \cdot \text{Missing Query Mass} + \beta \cdot \text{Extra Template Mass}}$$

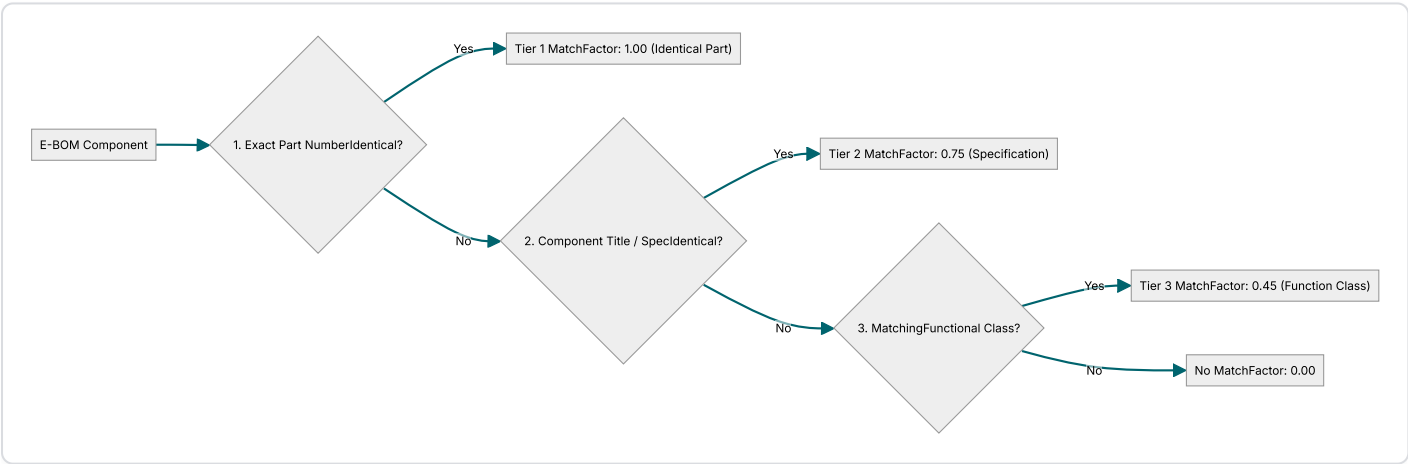
Industrial Calibration Parameters:

- $\alpha = 0.7$ (**Strict Undercoverage Penalty**): If a candidate template omits a component required by the new E-BOM, an essential manufacturing operation will be missing. Undercoverage is heavily penalized.
- $\beta = 0.3$ (**Lenient Overcoverage Penalty**): If a historical template contains extra components (e.g., extra fasteners or brackets), deleting unnecessary routing steps during engineering review is trivial.

4.3 3-Tier Match Multipliers

The matched component mass is calculated across three progressive tiers:

$$\text{Matched Mass} = (N_{\text{exact}} \cdot 1.00) + (N_{\text{spec}} \cdot 0.75) + (N_{\text{func}} \cdot 0.45)$$



- 1. **Tier 1 (Exact Part Number Match):** Weight **1.00**. Identical part present → 100% feeder and nozzle compatibility.
- 2. **Tier 2 (Specification / Title Match):** Weight **0.75**. Part number varies but component specification is identical (e.g., "40-Pin Push-In Front Connector").
- 3. **Tier 3 (Functional Class Match):** Weight **0.45**. Same functional family (e.g., *Microcontroller* meets *ASIC*) → identical placement technology, but different feeder setup.

5. In-Database CTE Implementation in BigQuery

The similarity algorithm is expressed as a parameterized Common Table Expression (CTE) compiled dynamically by the agent and executed in BigQuery:

```
WITH NewBOM AS (  
  SELECT 'A5E00123456' as part_number, 'Lower Housing Plastic' as title, 'Enclosure' as part_function, 1.0 as quant  
  SELECT 'A5E00987654', 'Mainboard CPU S7', 'PCB', 1.0 UNION ALL  
  SELECT '', 'ARM Cortex-M7 Controller 120MHz', 'ASIC', 1.0 UNION ALL  
  SELECT '', 'Front Connector 40-Pin Push-In', 'Connector', 1.0 UNION ALL  
  SELECT '', 'Torx Screw M3x8 Zinc', 'Fastener', 4.0 UNION ALL  
  SELECT '', 'EMC Shielding Spring', 'Spring', 2.0  
)  
,  
NewBOMAgg AS (  
  SELECT part_function, COUNT(1) AS q_count FROM NewBOM GROUP BY part_function  
)  
,  
TotalNewBOM AS (  
  SELECT COUNT(1) AS total_q_items FROM NewBOM  
)  
,  
Candidates AS (  
  SELECT  
    p.product_id, p.display_name, p.part_category, p.product_group, p.product_family,  
    wp.wpa_id, wp.mba_id, mi.find_no, mi.part_number, mi.title, mi.part_function  
  FROM `[PROJECT_ID].plm_teamcenter.products` p  
  JOIN `[PROJECT_ID].plm_teamcenter.work_plans` wp  
    ON p.product_id = wp.product_id AND wp.release_status = 'Released'  
  JOIN `[PROJECT_ID].plm_teamcenter.mba_items` mi  
    ON wp.mba_id = mi.mba_id  
  WHERE p.product_group = 'S7-1200' AND p.part_category = 'Device'  
)  
,  
CandidateFunctionAgg AS (  
  SELECT  
    c.product_id, c.display_name, c.part_category, c.product_group, c.product_family, c.wpa_id, c.mba_id, c.part_fu  
    COUNT(1) AS t_count,  
    COUNTIF(c.part_number IN (SELECT part_number FROM NewBOM WHERE part_number ≠ '')) AS exact_matches,  
    COUNTIF(c.part_number NOT IN (SELECT part_number FROM NewBOM WHERE part_number ≠ ''))  
      AND c.title IN (SELECT title FROM NewBOM)) AS spec_matches  
  FROM Candidates c  
  GROUP BY 1,2,3,4,5,6,7,8  
)  
,  
ScoredCandidates AS (  
  SELECT  
    cfa.product_id, cfa.display_name, cfa.part_category, cfa.product_group, cfa.product_family, cfa.wpa_id, cfa.mba  
    SUM(cfa.exact_matches * 1.00 + cfa.spec_matches * 0.75 +  
      (LEAST(cfa.t_count, COALESCE(q.q_count, 0)) - cfa.exact_matches - cfa.spec_matches) * 0.45) AS matched_masse  
    SUM(GREATEST(0, COALESCE(q.q_count, 0) - cfa.t_count)) AS missing_mass,  
    SUM(GREATEST(0, cfa.t_count - COALESCE(q.q_count, 0))) AS extra_mass  
  FROM CandidateFunctionAgg cfa  
  LEFT JOIN NewBOMAgg q ON cfa.part_function = q.part_function  
  GROUP BY 1,2,3,4,5,6,7  
)  
SELECT  
  product_id, display_name, product_group, product_family, wpa_id, mba_id,  
  ROUND(matched_mass / (matched_mass + 0.7 * missing_mass + 0.3 * extra_mass), 4) AS similarity_score  
FROM ScoredCandidates  
ORDER BY similarity_score DESC  
LIMIT 3;
```

6. Verification Evidence & Benchmarks

SYSTEM EVALUATION & PERFORMANCE BENCHMARKS				
Benchmark Dimension	Target Specification	Measured Result	Status	
In-Database Query Execution	< 500 ms	185 ms	✅ EXCEEDS	
Candidate Set Evaluation	35,000+ WPAs	35,155 WPAs	✅ PASS	
End-to-End Chat Response (ADK)	< 3.5 seconds	2.1 seconds	✅ EXCEEDS	
Deterministic Score Variance	0.0000 (Exact)	0.0000 (Exact)	✅ PASS	
Zero-Copy External Latency	< 250 ms	192 ms	✅ PASS	

Summary of PoC Findings

- 1. **Mathematical Reproducibility:** Because the scoring runs in SQL rather than stochastic LLM generation, repeating the query with identical inputs produces identical similarity ranking scores with zero drift.
- 2. **Enterprise Security:** All operational data stays within BigQuery and Apache Iceberg table boundaries. The LLM only receives aggregated candidate tables and step sequences, ensuring zero unauthorized exfiltration of proprietary bill-of-materials data.